2.3 Abstract Algebra II: Study Notes

Conventions: rings are commutative with 1 unless stated otherwise; ring homomorphisms send 1 to 1. F, K, L, E denote fields.

Part A: Rings

A1. Ideals and quotients (revision)

An *ideal* $I \subseteq R$ is an additive subgroup with $rI \subseteq I$ for all $r \in R$. The quotient R/I is a ring, and the isomorphism theorems hold as for groups: if $\varphi : R \to S$ is a homomorphism, $R/\ker\varphi \cong \operatorname{im}\varphi$.

- P is *prime* if $P \neq R$ and $ab \in P \Rightarrow a \in P$ or $b \in P$. Equivalently, R/P is an integral domain.
- M is *maximal* if $M \neq R$ and no ideal lies strictly between M and R. Equivalently, R/M is a field. Maximal ⇒ prime.
- By Zorn's lemma, every proper ideal is contained in a maximal ideal.
- **Chinese remainder theorem.** If $I_1, \dots, I_k$ are pairwise *comaximal* ($I_i + I_j = R$), then $R/(I_1 \cap \cdots \cap I_k) \cong \prod R/I_j$. For $R = \mathbb{Z}$: $\mathbb{Z}/mn \cong \mathbb{Z}/m \times \mathbb{Z}/n$ when $\gcd(m, n) = 1$.

A2. Euclidean domains, PIDs, UFDs

Let R be an integral domain.

- R is a *Euclidean domain* if there is $N : R \setminus \{0\} \to \mathbb{Z}_{\geq 0}$ such that for all a and $b \neq 0$ there are q, r with $a = qb + r$ and $r = 0$ or $N(r) < N(b)$. Examples: $\mathbb{Z}$, $F[x]$, $\mathbb{Z}[i]$ with $N(a + bi) = a^2 + b^2$.
- R is a *principal ideal domain (PID)* if every ideal is (a) for some a.
- R is a *unique factorisation domain (UFD)* if every non-zero non-unit is a product of irreducibles, uniquely up to order and units.

Theorem. Euclidean ⇒ PID ⇒ UFD.

Euclidean ⇒ PID: a non-zero ideal I is generated by an element of minimal norm (the remainder of any element on division by it lies in I and has smaller norm, so is 0).

PID ⇒ UFD (sketch): (i) In a PID, irreducible ⇒ prime: if p is irreducible, (p) is maximal among principal ideals, hence maximal, hence prime. (ii) Existence of factorisations: a PID is *Noetherian* (an ascending chain of ideals $(a_1) \subseteq (a_2) \subseteq \cdots$ has union an ideal (a), and a lies in some (a_n), so the chain stabilises); if some element had no factorisation, repeatedly splitting it would give a strictly ascending chain. (iii) Uniqueness follows from "irreducible ⇒ prime" exactly as in $\mathbb{Z}$.

Counterexamples.

- $\mathbb{Z}[\sqrt{-5}]$ is not a UFD: $6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$, and all four factors are irreducible and non-associate (use the norm $a^2 + 5b^2$: there is no element of norm 2 or 3).
- $\mathbb{Z}[x]$ and $F[x, y]$ are UFDs but not PIDs: $(2, x)$ and (x, y) are not principal.
- $\mathbb{Z}[(1 + \sqrt{-19})/2]$ is a PID but not Euclidean (statement; see DF 8.2).

In a PID, $\gcd(a, b)$ exists and generates (a, b), so it can be written $ra + sb$ (Bézout).

A3. Polynomial rings and irreducibility

Gauss's lemma. Let R be a UFD with fraction field F. If $f \in R[x]$ is reducible in $F[x]$, it is reducible in $R[x]$ (into factors of the same degrees as in $F[x]$, after rescaling). A primitive

polynomial (coefficients with gcd 1) is irreducible in $R[x]$ iff it is irreducible in $F[x]$. Consequence: **if R is a UFD, so is $R[x]$.** Hence $\mathbb{Z}[x_1, \dots, x_n]$ and $F[x_1, \dots, x_n]$ are UFDs.

Tests for irreducibility over $\mathbb{Q}$.

1. *Degree 2 or 3:* irreducible iff no root. Use the rational root test: a root p/q in lowest terms of $a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$ has $p \mid a_0$, $q \mid a_n$.
2. *Eisenstein:* if a prime p divides $a_0, \dots, a_{n-1}$, $p \nmid a_n$ and $p^2 \nmid a_0$, then f is irreducible over $\mathbb{Q}$. Often applied after a substitution $x \mapsto x + c$: e.g. $\Phi_p(x) = \frac{x^p - 1}{x - 1}$ becomes Eisenstein at p after $x \mapsto x + 1$.
3. *Reduction mod p:* if $p \nmid a_n$ and $\bar{f} \in \mathbb{F}_p[x]$ is irreducible, then f is irreducible over $\mathbb{Q}$. (The converse fails: $x^4 + 1$ is irreducible over $\mathbb{Q}$ but reducible mod every prime.)

Part B: Modules

B1. Basics

An *R-module* is an abelian group M with a scalar multiplication $R \times M \to M$ satisfying the vector space axioms. Examples:

- R-modules for $R = F$ a field are vector spaces.
- $\mathbb{Z}$-modules are exactly abelian groups.
- An $F[x]$-module is a vector space V together with a linear map $T : V \to V$ (where x acts as T). Submodules are T-invariant subspaces. This dictionary is how module theory proves canonical form theorems.
- Ideals of R are the submodules of R.

Submodules, quotients, homomorphisms, kernels, and the isomorphism theorems work as for groups. The *direct sum* $M \oplus N$ is the product with componentwise operations.

A module is *free* if it has a *basis* (a linearly independent spanning set); then $M \cong R^{(I)}$. For commutative R, all bases of a free module have the same cardinality (the *rank*). Over a non-field, many modules are not free: $\mathbb{Z}/2$ is not a free $\mathbb{Z}$-module, and neither is $\mathbb{Q}$.

$m \in M$ is a *torsion* element if $rm = 0$ for some non-zero $r \in R$. Over a domain, the torsion elements form a submodule $\mathrm{Tor}(M)$.

Pitfalls (compared with vector spaces). A linearly independent set need not extend to a basis ($\{2\}$ in $\mathbb{Z}$); a spanning set need not contain a basis ($\{2, 3\}$ spans $\mathbb{Z}$); a submodule need not have a complement ($2\mathbb{Z} \subseteq \mathbb{Z}$); a submodule of a finitely generated module need not be finitely generated (over non-Noetherian rings).

B2. Submodules of free modules over a PID

Theorem. Let R be a PID. Every submodule N of R^n is free of rank $\leq n$.

Proof. Induction on n. For $n = 1$, N is an ideal (a), which is 0 or free on a (as R is a domain). For $n > 1$, let $\pi : R^n \to R$ be the last coordinate. Then $\pi(N)$ is an ideal (a), and $\ker(\pi|_N) = N \cap R^{n-1}$ is free of rank $\leq n - 1$ by induction. If $a = 0$, done. Otherwise pick $y \in N$ with $\pi(y) = a$; then $N = (N \cap R^{n-1}) \oplus Ry$, so N is free of rank $\leq n$. $\square$

B3. Smith normal form and the structure theorem

Smith normal form. Let R be a PID (for computation, think Euclidean) and A an $m \times n$ matrix over R. There are invertible matrices $P \in GL_m(R)$, $Q \in GL_n(R)$ with

$$PAQ = \mathrm{diag}(d_1, d_2, \dots, d_r, 0, \dots, 0), \qquad d_1 \mid d_2 \mid \cdots \mid d_r, \quad d_i \neq 0.$$

The d_i are unique up to units: $d_1 \cdots d_k$ is the gcd of the $k \times k$ minors of A.

Algorithm (Euclidean case). Using row and column swaps, adding multiples of one row (column) to another, and multiplying by units: move an entry of minimal norm to position $(1,1)$; use division with remainder to clear the first row and column; if a remainder is non-zero it has smaller norm, so move it to $(1,1)$ and repeat. Once row and column 1 are clear, if d_1 does not divide some entry b, add that row to row 1 and repeat (this lowers the norm of the $(1,1)$ entry). Then recurse on the lower-right block.

Structure theorem (existence). Let R be a PID and M a finitely generated R-module. Then

$$M \cong R^r \oplus R/(d_1) \oplus \cdots \oplus R/(d_k), \qquad d_1 \mid d_2 \mid \cdots \mid d_k,\ d_i \text{ non-zero non-units.}$$

Proof. Choose generators $m_1, \dots, m_n$, giving a surjection $\varphi : R^n \to M$. By B2, $\ker \varphi$ is free of rank $s \le n$; choosing a basis gives an $n \times s$ matrix A whose columns span $\ker \varphi$, so $M \cong R^n / AR^s$. Changing bases of R^n and R^s replaces A by PAQ, so we may assume A is in Smith normal form. Then $R^n/AR^s \cong \bigoplus_i R/(d_i) \oplus R^{n-s}$. Drop factors $R/(d_i)$ with d_i a unit (they are zero). □

Uniqueness. The rank r and the ideals (d_i) (the *invariant factors*) are uniquely determined by M. (One proof: the rank is $\dim_F(F \otimes M)$ for F the fraction field; for the torsion part, the number of d_i divisible by a power p^j is read off from $\dim_{R/(p)} p^{j-1}M/p^jM$.)

Elementary divisor form. By the Chinese remainder theorem, each $R/(d)$ with $d = u\, p_1^{a_1} \cdots p_t^{a_t}$ splits as $\bigoplus R/(p_i^{a_i})$. So

$$M \cong R^r \oplus \bigoplus_i R/(p_i^{a_i}),$$

with the prime powers $p_i^{a_i}$ (the *elementary divisors*) unique up to order and units. To convert back: for each prime, list its powers in decreasing order; the largest invariant factor is the product of the largest power of each prime, and so on.

B4. Application 1: finitely generated abelian groups

With $R = \mathbb{Z}$: every finitely generated abelian group is $\mathbb{Z}^r \oplus \mathbb{Z}/d_1 \oplus \cdots \oplus \mathbb{Z}/d_k$ with $1 < d_1 \mid \cdots \mid d_k$, uniquely. The number of abelian groups of order $p_1^{a_1} \cdots p_t^{a_t}$ is $\prod p(a_i)$, where $p(a)$ is the number of partitions of a.

Example. Order $72 = 2^3 \cdot 3^2$: $p(3)p(2) = 3 \cdot 2 = 6$ groups.

Example. If $G = \mathbb{Z}^3/A\mathbb{Z}^3$ and A has Smith normal form $\operatorname{diag}(2,6,12)$, then $G \cong \mathbb{Z}/2 \oplus \mathbb{Z}/6 \oplus \mathbb{Z}/12$, of order $|\det A| = 144$.

B5. Application 2: canonical forms

Let V be an n-dimensional vector space over F and $T : V \to V$ linear. Make V an $F[x]$-module via T. It is finitely generated and torsion (by Cayley–Hamilton, or simply because V is finite-dimensional so $I, T, T^2, \dots$ are dependent). So

$$V \cong F[x]/(a_1) \oplus \cdots \oplus F[x]/(a_k), \qquad a_1 \mid \cdots \mid a_k,\ a_i \text{ monic of degree} \ge 1.$$

- On $F[x]/(a)$ with $a = x^d + c_{d-1}x^{d-1} + \cdots + c_0$, multiplication by x in the basis $1, x, \dots, x^{d-1}$ is the **companion matrix** $C(a)$ (ones on the subdiagonal, last column $-c_0, \dots, -c_{d-1}$).
- **Rational canonical form:** T has matrix $C(a_1) \oplus \cdots \oplus C(a_k)$. It exists over any field and is unique.
- The **minimal polynomial** of T is a_k; the **characteristic polynomial** is $a_1 \cdots a_k$. This proves Cayley–Hamilton: $m_T \mid \chi_T$.

- **Jordan canonical form:** if χ_T splits over F (e.g. $F = \mathbb{C}$), the elementary divisors are $(x-\lambda)^e$, and on $F[x]/((x-\lambda)^e)$ with basis $(x-\lambda)^{e-1}, \dots, (x-\lambda), 1$ the matrix is a Jordan block $J_e(\lambda)$. So T is similar to a direct sum of Jordan blocks, unique up to order.
- Two matrices are similar iff they have the same invariant factors (equivalently, same rational canonical form). They are similar over F iff they are similar over any extension field $K \supseteq F$ (invariant factors can be computed by Smith normal form of $xI - A$ over $F[x]$, which does not change when extending the field).

Reading off Jordan form data. For eigenvalue λ:

- the exponent of $(x-\lambda)$ in χ_T = total size of the λ-blocks (algebraic multiplicity);
- the exponent in m_T = size of the largest λ-block;
- $\dim\ker(T-\lambda)$ = number of λ-blocks (geometric multiplicity);
- $\dim\ker(T-\lambda)^j - \dim\ker(T-\lambda)^{j-1}$ = number of λ-blocks of size $\geq j$.

T is diagonalisable iff m_T splits with distinct roots.

Part C: Fields

C1. Extensions and degree

A *field extension* K/F is a field K containing F. K is an F-vector space; its dimension $[K : F]$ is the *degree*.

Tower law. If $F \subseteq K \subseteq L$, then $[L : F] = [L : K][K : F]$ (if $\{\alpha_i\}$ is a basis of K/F and $\{\beta_j\}$ of L/K, then $\{\alpha_i\beta_j\}$ is a basis of L/F).

$\alpha \in K$ is *algebraic* over F if $f(\alpha) = 0$ for some non-zero $f \in F[x]$; otherwise *transcendental*. The *minimal polynomial* m_α is the monic generator of the kernel of $F[x] \to K$, $f \mapsto f(\alpha)$. It is irreducible, and

$$F(\alpha) = F[\alpha] \cong F[x]/(m_\alpha), \qquad [F(\alpha) : F] = \deg m_\alpha.$$

A basis is $1, \alpha, \dots, \alpha^{d-1}$. Conversely, for any irreducible $f \in F[x]$, $F[x]/(f)$ is a field containing a root of f (**Kronecker's construction**).

- $[K : F] < \infty \Rightarrow K/F$ algebraic. $F(\alpha_1, \dots, \alpha_n)$ with each α_i algebraic is finite. Algebraic over algebraic is algebraic. The algebraic elements of K over F form a subfield.
- Example: $\overline{\mathbb{Q}}$ (algebraic numbers in $\mathbb{C}$) is algebraic over $\mathbb{Q}$ of infinite degree.

Ruler-and-compass constructions. If $\alpha \in \mathbb{R}$ is constructible from $0, 1$, then α lies in a tower $\mathbb{Q} = F_0 \subseteq F_1 \subseteq \cdots \subseteq F_k$ with $[F_i : F_{i-1}] = 2$, so $[\mathbb{Q}(\alpha) : \mathbb{Q}]$ is a power of 2. Hence: $\sqrt[3]{2}$ is not constructible (no duplicating the cube); $\cos 20^\circ$ has degree 3 (no trisecting 60°); π is transcendental (Lindemann), so the circle cannot be squared. The regular n-gon is constructible iff $\varphi(n)$ is a power of 2 (Gauss–Wantzel; the "if" direction uses Galois theory).

C2. Splitting fields and algebraic closure

A *splitting field* of $f \in F[x]$ is an extension K in which f factors into linear factors and which is generated by the roots of f.

Theorem. Splitting fields exist (adjoin roots one at a time using Kronecker's construction), $[K : F] \leq n!$ for $\deg f = n$, and any two splitting fields of f over F are isomorphic over F.

The uniqueness rests on the **extension lemma**: if $\sigma : F \to F'$ is an isomorphism, $f \in F[x]$, K a splitting field of f and K' a splitting field of σf, then σ extends to an isomorphism $K \to K'$. The number of extensions is at most $[K : F]$, with equality iff every irreducible factor of f has distinct roots (separability). *Proof idea:* if α is a root of an irreducible factor g of f, then σ extends to $F(\alpha) \cong F[x]/(g) \to F'[x]/(\sigma g) \cong F'(\alpha')$ for each root α' of σg in K'; induct on $[K : F]$.

Examples. $x^3 - 2$ over $\mathbb{Q}$: splitting field $\mathbb{Q}(\sqrt[3]{2}, \omega)$, $\omega = e^{2\pi i/3}$, degree 6. $x^p - 1$: splitting field $\mathbb{Q}(\zeta_p)$, degree $p-1$. $x^4 - 2$: $\mathbb{Q}(\sqrt[4]{2}, i)$, degree 8.

Algebraic closure. A field Ω is *algebraically closed* if every non-constant polynomial in $\Omega[x]$ has a root in Ω. An *algebraic closure* $\overline{F}$ of F is an algebraic extension that is algebraically closed. Every field has an algebraic closure, unique up to (non-unique) isomorphism over F (Zorn's lemma; Milne Ch. 6). $\mathbb{C}$ is algebraically closed (fundamental theorem of algebra; see module 2.2), so $\overline{\mathbb{Q}}$ = algebraic numbers in $\mathbb{C}$.

C3. Separability

$f \in F[x]$ is *separable* if it has distinct roots in a splitting field. **Test:** f is separable iff $\gcd(f, f') = 1$.

- If f is irreducible, then $\gcd(f, f')$ is 1 or f; the latter happens iff $f' = 0$. In characteristic 0, $f' = 0$ only for constants, so **every irreducible polynomial is separable**. In characteristic p, $f' = 0$ iff $f(x) = g(x^p)$.
- F is *perfect* if every irreducible polynomial over F is separable. Fields of characteristic 0 and finite fields are perfect. (Over $\mathbb{F}_q$, $g(x^p) = h(x)^p$ by surjectivity of Frobenius.)
- Non-example: over $F = \mathbb{F}_p(t)$, $x^p - t$ is irreducible (Eisenstein at the prime t of $\mathbb{F}_p[t]$) but equals $(x - t^{1/p})^p$ in its splitting field.

An algebraic extension K/F is *separable* if the minimal polynomial of every element is separable.

Primitive element theorem. A finite separable extension K/F is simple: $K = F(\gamma)$ for some γ. (For infinite F: show $F(\alpha, \beta) = F(\alpha + c\beta)$ for all but finitely many $c \in F$. For finite F: $K^\times$ is cyclic.) Example: $\mathbb{Q}(\sqrt{2}, \sqrt{3}) = \mathbb{Q}(\sqrt{2} + \sqrt{3})$.

C4. Finite fields

Let F be a finite field. Its prime subfield is $\mathbb{F}_p$ for $p = \operatorname{char} F$, so $|F| = p^n$ where $n = [F : \mathbb{F}_p]$.

Theorem.

1. For each prime power $q = p^n$ there is a field $\mathbb{F}_q$ with q elements: the splitting field of $x^q - x$ over $\mathbb{F}_p$, which consists exactly of the roots of $x^q - x$ (they form a field because $(a+b)^p = a^p + b^p$, and they are distinct because $(x^q - x)' = -1$).
2. Any field with q elements is a splitting field of $x^q - x$ (each element satisfies $a^q = a$ by Lagrange on $F^\times$), hence isomorphic to $\mathbb{F}_q$.
3. $\mathbb{F}_q^\times$ is cyclic. (A finite subgroup of the multiplicative group of any field is cyclic.)
4. $\mathbb{F}_{p^m} \subseteq \mathbb{F}_{p^n}$ iff $m \mid n$, and then the subfield is unique.
5. $\mathbb{F}_{p^n}/\mathbb{F}_p$ is Galois with cyclic group of order n generated by the **Frobenius** $\phi(a) = a^p$.
6. $x^{p^n} - x = \prod$ (all monic irreducible polynomials over $\mathbb{F}_p$ of degree dividing n). Counting degrees gives $p^n = \sum_{d \mid n} d\, N_d$, where N_d = number of monic irreducibles of degree d.

Part D: Galois theory

D1. Galois extensions

For an extension K/F, $\mathrm{Aut}(K/F)$ is the group of field automorphisms of K fixing F pointwise. For a group H of automorphisms of K, K^H is the fixed field.

Basic bound. If K/F is finite, $|\mathrm{Aut}(K/F)| \le [K : F]$.

Definition/Theorem. For a finite extension K/F the following are equivalent; if they hold, K/F is *Galois* and $\mathrm{Gal}(K/F) := \mathrm{Aut}(K/F)$:

1. $|\mathrm{Aut}(K/F)| = [K : F]$;
2. $F = K^{\mathrm{Aut}(K/F)}$;
3. K is the splitting field of a separable polynomial over F;
4. K/F is **normal** (every irreducible polynomial in $F[x]$ with a root in K splits in K) and separable.

Artin's theorem. If G is a finite group of automorphisms of a field K, then $[K : K^G] = |G|$ and K/K^G is Galois with group G.

Proof idea. For $\alpha \in K$, the polynomial $\prod_{\beta \in G\alpha}(x - \beta)$ is fixed by G, so has coefficients in K^G; it is separable of degree $\leq |G|$. So every element of K has degree $\leq |G|$ over K^G, and K/K^G is separable; by the primitive element theorem, $[K : K^G] \leq |G|$. Combined with $|G| \leq |\mathrm{Aut}(K/K^G)| \leq [K : K^G]$, equality holds.

Examples.

- $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$: Galois, group $\mathbb{Z}/2$.
- $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$: **not** Galois. Its only automorphism is the identity, because the other roots of x^3-2 are not real. Not normal.
- $\mathbb{F}_p(t)/\mathbb{F}_p(t^p)$: not Galois (not separable).

D2. The fundamental theorem

Fundamental theorem of Galois theory. Let K/F be finite Galois with group G. Then

$$H \longmapsto K^H, \qquad E \longmapsto \mathrm{Gal}(K/E)$$

are mutually inverse inclusion-reversing bijections between subgroups $H \leq G$ and intermediate fields $F \subseteq E \subseteq K$. Moreover:

1. $[K : K^H] = |H|$ and $[K^H : F] = [G : H]$.
2. K/E is always Galois. E/F is Galois iff $\mathrm{Gal}(K/E)$ is normal in G, and then $\mathrm{Gal}(E/F) \cong G/\mathrm{Gal}(K/E)$ (by restriction).
3. For $\sigma \in G$, $\sigma(K^H) = K^{\sigma H \sigma^{-1}}$: conjugate subgroups $\leftrightarrow$ conjugate fields.
4. Joins and meets correspond: $K^{H_1 \cap H_2} = K^{H_1}K^{H_2}$ (compositum), $K^{\langle H_1, H_2\rangle} = K^{H_1} \cap K^{H_2}$.

Proof sketch. K/E is Galois because K is the splitting field over E of the same separable polynomial. So $E = K^{\mathrm{Gal}(K/E)}$. Artin gives $\mathrm{Gal}(K/K^H) = H$ (since $|\mathrm{Gal}(K/K^H)| = [K : K^H] = |H|$ and it contains H). So the maps are inverse. For (2): $\sigma(E) = E$ for all σ iff $\mathrm{Gal}(K/E)$ is normal (by 3); then restriction $G \to \mathrm{Aut}(E/F)$ has kernel $\mathrm{Gal}(K/E)$, and counting gives the isomorphism.

Worked example: $x^3 - 2$ over $\mathbb{Q}$. $K = \mathbb{Q}(\alpha, \omega)$ with $\alpha = \sqrt[3]{2}$, $\omega = e^{2\pi i/3}$, $[K : \mathbb{Q}] = 6$. G acts faithfully on the three roots $\alpha, \alpha\omega, \alpha\omega^2$, so $G \leq S_3$; with order 6, $G \cong S_3$. Subgroups and fixed fields:

Subgroup	Order	Fixed field	Normal?
1	1	K	yes
$\langle\tau\rangle$, τ = complex conjugation (fixes α)	2	$\mathbb{Q}(\alpha)$	no
transposition fixing $\alpha\omega$	2	$\mathbb{Q}(\alpha\omega)$	no

Subgroup	Order	Fixed field	Normal?
transposition fixing $\alpha\omega^2$	2	$\mathbb{Q}(\alpha\omega^2)$	no
$A_3 = \langle\sigma\rangle$, $\sigma : \alpha \mapsto \alpha\omega,\ \omega \mapsto \omega$	3	$\mathbb{Q}(\omega) = \mathbb{Q}(\sqrt{-3})$	yes
S_3	6	$\mathbb{Q}$	yes

Worked example: cyclotomic fields. $\Phi_n(x) = \prod_{\gcd(k,n)=1}(x - \zeta_n^k)$ has integer coefficients and is irreducible over $\mathbb{Q}$ (for $n = p$ prime: Eisenstein after $x \mapsto x + 1$; general n: DF 13.6). $\mathbb{Q}(\zeta_n)$ is the splitting field of $x^n - 1$, and $\sigma_a : \zeta_n \mapsto \zeta_n^a$ gives

$$\mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n)^\times,$$

an abelian group. Every subfield is Galois over $\mathbb{Q}$. (Kronecker–Weber, statement: every abelian extension of $\mathbb{Q}$ lies in some $\mathbb{Q}(\zeta_n)$.)

Galois groups of polynomials. $\mathrm{Gal}(f)$ acts faithfully on the n roots, so embeds in S_n; it is transitive iff f is irreducible (for separable f). With the discriminant $\Delta = \prod_{i<j}(\alpha_i - \alpha_j)^2 \in F$: $\mathrm{Gal}(f) \le A_n$ iff $\sqrt{\Delta} \in F$ (char $\neq 2$). For an irreducible cubic: Gal $= A_3$ if Δ is a square, else S_3. For $x^3 + px + q$, $\Delta = -4p^3 - 27q^2$.

D3. Solvability by radicals

A group G is *solvable* if it has a chain $1 = G_0 \trianglelefteq G_1 \trianglelefteq \cdots \trianglelefteq G_k = G$ with each G_i/G_{i-1} abelian. Subgroups and quotients of solvable groups are solvable, and an extension of solvable by solvable is solvable. S_n is solvable for $n \le 4$, and not for $n \ge 5$ because A_n is simple and non-abelian for $n \ge 5$.

A finite extension K/F is *radical* if there is a tower $F = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_m \supseteq K$ with $K_i = K_{i-1}(\alpha_i)$ and $\alpha_i^{n_i} \in K_{i-1}$. f is *solvable by radicals* if its splitting field lies in a radical extension.

Key ingredients (characteristic 0).

1. If F contains a primitive n-th root of unity and $a \in F$, then $F(\sqrt[n]{a})/F$ is Galois with cyclic group (of order dividing n). (**Kummer**; Problem 22.)
2. $F(\zeta_n)/F$ is Galois with abelian group (a subgroup of $(\mathbb{Z}/n)^\times$).
3. Conversely (Kummer, with Hilbert 90): if F contains a primitive n-th root of unity and K/F is cyclic of degree n, then $K = F(\sqrt[n]{a})$.

Theorem (Galois). In characteristic 0, $f \in F[x]$ is solvable by radicals iff $\mathrm{Gal}(f)$ is solvable.

Proof of "⇒" (sketch). Enlarge the radical tower: adjoin a primitive N-th root of unity first (N = product of the n_i), and take the Galois closure (still radical). Each step is then abelian (by 1 and 2), so the Galois group of the top field over F is solvable. $\mathrm{Gal}(f)$ is a quotient of it, hence solvable.

Insolubility of the quintic. Let $f \in \mathbb{Q}[x]$ be irreducible of prime degree p with exactly $p - 2$ real roots. Then $\mathrm{Gal}(f) \cong S_p$.

Proof. p divides $|G|$ (since $\mathbb{Q}(\alpha)$ has degree p and lies in the splitting field), so by Cauchy G contains an element of order p, which in S_p is a p-cycle. Complex conjugation restricts to an element of G swapping the two non-real roots and fixing the others: a transposition. A p-cycle and a transposition generate S_p (p prime). □

Example: $x^5 - 6x + 3$ (Eisenstein at 3; three real roots by the intermediate value theorem and Rolle). Its Galois group S_5 is not solvable, so its roots cannot be expressed by radicals. Note that the general quintic $x^5 + a_4x^4 + \cdots + a_0$ over $\mathbb{Q}(a_0, \ldots, a_4)$ has group S_5 too (Abel–Ruffini).

Common pitfalls

1. "Irreducible" and "prime" coincide in UFDs but not in general (2 in $\mathbb{Z}[\sqrt{-5}]$ is irreducible, not prime).
2. A polynomial can be irreducible over $\mathbb{Q}$ but have a reducible reduction mod every p ($x^4 + 1$). Reduction mod p only proves irreducibility, never reducibility.
3. $[F(\alpha, \beta) : F]$ is at most $[F(\alpha) : F]\,[F(\beta) : F]$ but need not equal it: $\alpha = \sqrt[3]{2}$, $\beta = \omega\sqrt[3]{2}$ give $\mathbb{Q}(\alpha, \beta) = \mathbb{Q}(\sqrt[3]{2}, \omega)$ of degree 6, not 9. (Equality does hold when the two degrees are coprime.)
4. $\mathrm{Aut}(K/F)$ is always defined, but the Galois correspondence needs K/F Galois.
5. Normality is not transitive: $\mathbb{Q} \subseteq \mathbb{Q}(\sqrt{2}) \subseteq \mathbb{Q}(\sqrt[4]{2})$, each step degree 2 (hence normal), but $\mathbb{Q}(\sqrt[4]{2})/\mathbb{Q}$ is not normal.
6. Invariant factors must divide each other; elementary divisors are prime powers. Do not mix the two lists.
7. Jordan form needs the characteristic polynomial to split; rational canonical form does not.

Connections forward

- **3.3 Graduate algebra:** tensor products, exact sequences, Noetherian rings and the Hilbert basis theorem, Nullstellensatz, localisation, Dedekind domains, representation theory of finite groups, infinite Galois theory.
- **3.4 Algebraic topology:** homology groups of finite CW complexes are finitely generated abelian groups; Smith normal form computes simplicial homology; the classification of covering spaces is a Galois correspondence (π_1 plays the role of the Galois group).
- **3.5 Smooth manifolds:** exterior algebra of a vector space; tensors.
- **Electives:** algebraic number theory (rings of integers, Dedekind domains, Galois action on primes), algebraic geometry (function fields, transcendence degree), commutative algebra.